ReturnLens

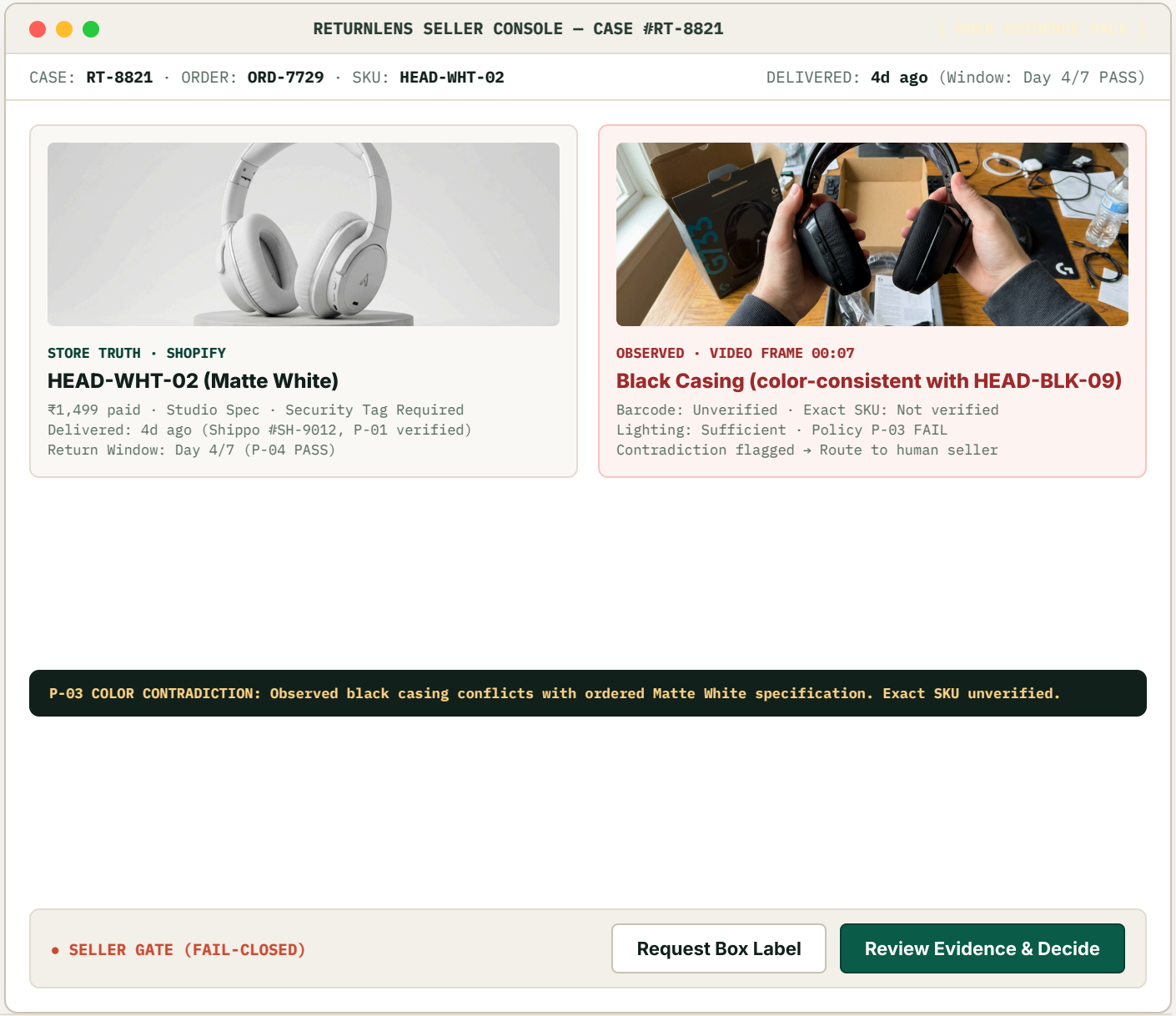
Every return decision, backed by evidence.

ReturnLens unifies unboxing video evidence, store truth, and policy checks into an evidence-backed triage console for solo D2C merchants.

CORE ARCHITECTURAL THESIS

“Models observe. Code verifies. Seller authorizes.”

Scope: Triage automation · Fail-closed gate · Never auto-refunds
Guardrail: Server cap ≤ min(₹4,999, remaining, paid) · Hash-bound



THE PROBLEM · SOLO MERCHANT BOTTLENECK · 4 SOURCES

One return. Four disconnected systems. Zero shared record.

Solo merchants cross-reference tabs, video footage, and policy rules manually before moving money.

SOURCE 01 BUYER MESSAGE
Unstructured text & unverified unboxing video in chat

SOURCE 02 ORDER RECORD
Shopify manifest: HEAD-WHT-02, Matte White, ₹1,499

SOURCE 03 COURIER STATUS
Shippo scan: delivered 4 days ago (P-01 verified, Day 4 of 7)

SOURCE 04 RETURN POLICY
Store rules: 7-day window (P-04), packaging tag required (P-02)

THE CRITICAL INFORMATION GAP

No auditable evidence record. One solo seller making high-stakes financial decisions across browser tabs.

Without an auditable evidence pack, the merchant either refunds blind (losing COGS and freight) or disputes unfairly (destroying customer trust and merchant ratings).

INDUSTRY FRAUD BASELINE (RIGOROUS PROVENANCE · DISTINCT SCOPES)

- 9.0% — Return Fraud (NRF/Appriss 2024: strict criminal fraud: unboxing swaps, empty boxes).
- 15.14% — Fraud + Policy Abuse (Appriss/Deloitte 2024: criminal fraud + policy abuse like wardrobing).

SOLO MERCHANT ASYMMETRIC RISK

- Wrong refund = Lost product COGS + 2-way shipping costs (unrecoverable margin loss).
- Unfair denial = Ruined customer trust, chargeback penalties & degraded seller account health.

STAR DEMO · CASE B · CORE BENCHMARK

The order says white. The video shows black.

ReturnLens flags the contradiction — the seller decides.

ORDER MANIFEST · STORE CATALOG TRUTH (#ORD-7729)



SKU: HEAD-WHT-02 · Spec: Matte White Casing

Price: ₹1,499 paid · Delivered: 4d ago (Shippo #SH-9012)
Return Window P-04 (7-Day Policy): PASS (Day 4 of 7)

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MOCK EVIDENCE PACK · VIDEO CITATION (FRAME 00:07)



Observed: Black Casing (color-consistent with HEAD-BLK-09)

Exact SKU: Not verified (Barcode unverified) · Policy P-03 FAIL
Lighting: Sufficient · REC 00:07 / 00:24 (Mock Keyframe)

POLICY P-03 RESULT: ESCALATE TO SELLER GATE (FAIL-CLOSED)

Guardrail: 0 < amt ≤ min(₹4,999, rem_ref, paid) · Prototype simulation

1. Accept as warehouse mis-shipment
2. Request photo of outside-box shipping label
3. Hold return for physical parcel inspection

SYSTEM ARCHITECTURE · FAIL-CLOSED SAFETY GATES

Models observe. Code verifies. Seller authorizes.

Exactly 3 model calls per triage pass. Weak evidence triggers stateful clarification. Zero automated refunds.

1. Planner	2. Tools & Vision	3. Policy Engine	4. Critic	5. Seller Gate	6. Audit Ledger
Model Call 1	Model Call 2	Deterministic Code	Model Call 3	Human Merchant	SQLite / JSON
Proposes typed JSON actions	FFmpeg keyframes + vision (fallible)	Evaluates rules P-01–P-08	Verifies cited timestamps	Fail-closed 1-click auth	Append-only signed audit

CALL BUDGET & MODEL RESTRICTION

3 Calls Per Triage Pass

1. Planner proposes typed tool actions.
2. Vision model inspects keyframes (fallible observation).
3. Critic evaluates citations against policy.

Case C re-upload triggers Pass 2 (max 6 calls total). Models never access payment APIs directly.

STATEFUL RESUMPTION (CASE C)

Clarification & Re-Film Loop

Low-light video (Policy P-06 FAIL, brightness score: 28/255 < threshold 50/255) triggers ask_for_evidence().

State saves to WAITING_FOR_EVIDENCE → Resumes on upload.

TERMINATION RULE: After 1 retry, escalates to seller with both attempts logged (never auto-denies).

ZERO-LLM FAST PATH

Zero-LLM Order Route

For routine "Where is my order?" inquiries, deterministic code regex routes directly to APIs:

Store API → Courier API → Status Reply

Zero LLM tokens · Zero vision overhead · <500ms response.

FAIL-CLOSED FINANCIAL GATE: requires_seller_approval: true · Approval binds case + evidence hash + policy version + nonce.

Server Cap: 0 < amount <= min(₹4,999, remaining_refundable, order_paid) · Idempotency key hash(order_id, return_id, nonce) enforced.

DEMO SCENARIOS · BEHAVIORAL MATRIX

Different evidence. Different action.

Four distinct triage behaviors across reproducible test fixtures. Zero automated policy approvals.

01 · PREPARE REVIEW

CASE A · CONSISTENT

EVIDENCE PROFILE:
Manifest matches catalog
Packaging tag verified (P-02)
Cosmetic defect observed
Within 7-day window (P-04)

POLICY ENGINE ACTION:
Drafts return shipping label & refund draft in pending_return_receipt.

Funds release only upon physical warehouse parcel receipt.

02 · ESCALATE SELLER

CASE B · CONTRADICTION

EVIDENCE PROFILE:
Ordered Matte White
Observed Black casing (00:07)
Exact SKU unverified
Barcode unverified

POLICY ENGINE ACTION:
Policy P-03 FAIL blocks automation. Inhibits label & refund draft.

Surfaces 3 seller options: mis-shipment / box label / inspection.

03 · CLARIFY LOOP

CASE C · LOW LIGHT

EVIDENCE PROFILE:
4s low-light unboxing video
Brightness score: 28/255 < 50/255
Barcode obscured
Only 2 usable frames

POLICY ENGINE ACTION:
Policy P-06 FAIL triggers ask_for_evidence().

Saves to WAITING_FOR_EVIDENCE; resumes on upload. Max 1 retry.

04 · INSPECT CONTINUITY

CASE D · CONTINUOUS

EVIDENCE PROFILE:
Single unbroken unboxing take
Sealed parcel to product display
Continuity inspection
Flags abrupt transitions

POLICY ENGINE ACTION:
Inspects sequence continuity and flags abrupt cuts.

Basic continuity in prototype; advanced forensics is cut first.

24-HOUR PROTOTYPE PLAN · TEAM ROOKIE (3 MEMBERS)

A credible prototype in 24 hours.

Strict scope boundaries and single-threaded ownership make the prototype buildable in one day.

00 – 06h · PHASE 01

Foundation

Mock Store/Shippo APIs · LangGraph state schema · Zero-LLM fast path

06 – 12h · PHASE 02

Evidence Pipeline

FFmpeg ~1fps extractor · Vision observation schema · Seller Console UI

12 – 18h · PHASE 03

Integration

Case C pause/resume loop · 4 test fixtures (A–D) · Adversarial critic review

18 – 24h · PHASE 04

Validation & Safety

Fail-closed authorization tests · 10-test smoke suite · Video demo recording

PERSON A · BACKEND, POLICY & AUTH

State Machine & Policy Engine

Stack: LangGraph · Pydantic · SQLite · SHA256

- Policy P-01 to P-08: Deterministic code rules (window, SKU, tag, refund cap)
- Zero-LLM Fast Path: Deterministic regex routing for order status queries
- Audit Ledger: Append-only JSON log & signed approval binding
- Financial Cap: Server calculation min(₹4,999, remaining, paid)

PERSON B · VISION & CONSOLE

Video Pipeline & Seller UI

Stack: FFmpeg · Hosted vision model (unpinned) · HTML/CSS/SVG

- Keyframe Pipeline: Uniform ~1fps sampling & brightness scoring (threshold 50/255)
- Structured Vision: Observation schema with timestamp citation metadata
- Adversarial Critic: Cross-checks citations against keyframe timestamps
- Seller Console: 1-click decision console with mock evidence pack

PERSON C · FIXTURES & QA

Scenarios, Lifecycle & QA

Stack: Pytest · Mock Fixtures · Automated Smoke QA

- Benchmark Cases: Staged fixtures for 4 demo cases (A–D)
- Case C Lifecycle: Re-film loop in WAITING_FOR_EVIDENCE (1 retry max)
- Fail-Closed Testing: Verifies zero automated refunds & approval gates
- Automated QA: 10-test smoke suite, boundary verification & demo recording

MUST DELIVER (CORE PROTOTYPE)

Mock APIs · Policy engine · 4 benchmark cases (A–D) · Approval binding · Seller gate · Zero-LLM fast path

CUT FIRST IF BEHIND

Advanced video forensics (Case D stretch) · WhatsApp bot · Exchanges · Multi-seller auth · Audio analysis · Public cloud deployment

PROTOTYPE SCOPE LOCKED

100% reproducible offline build · Zero external network dependencies